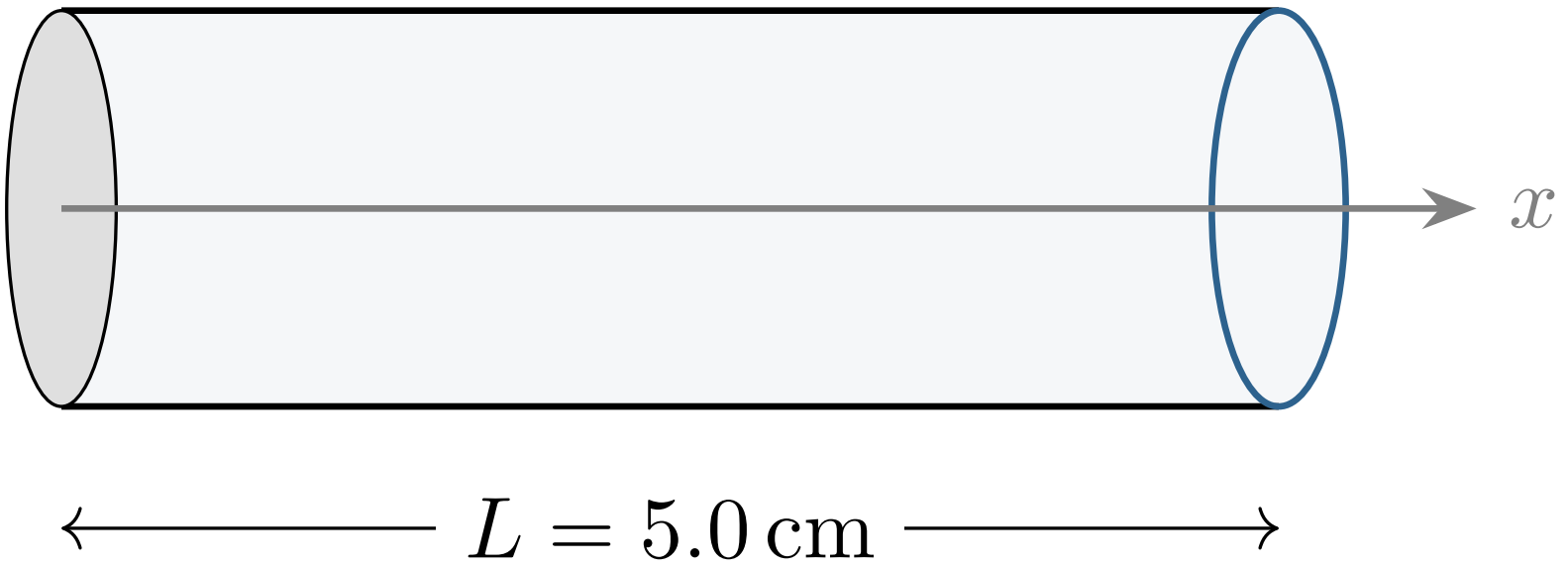
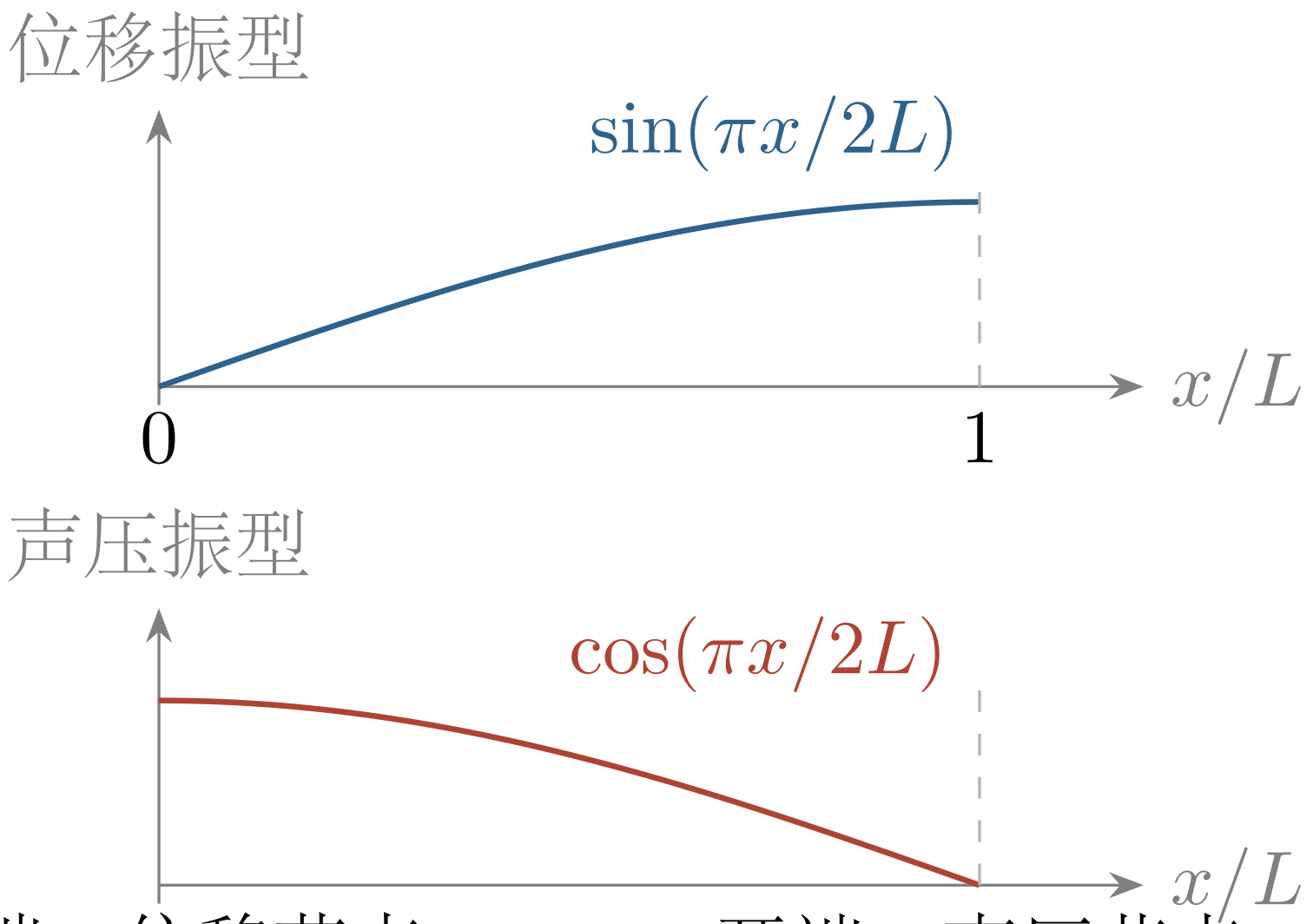
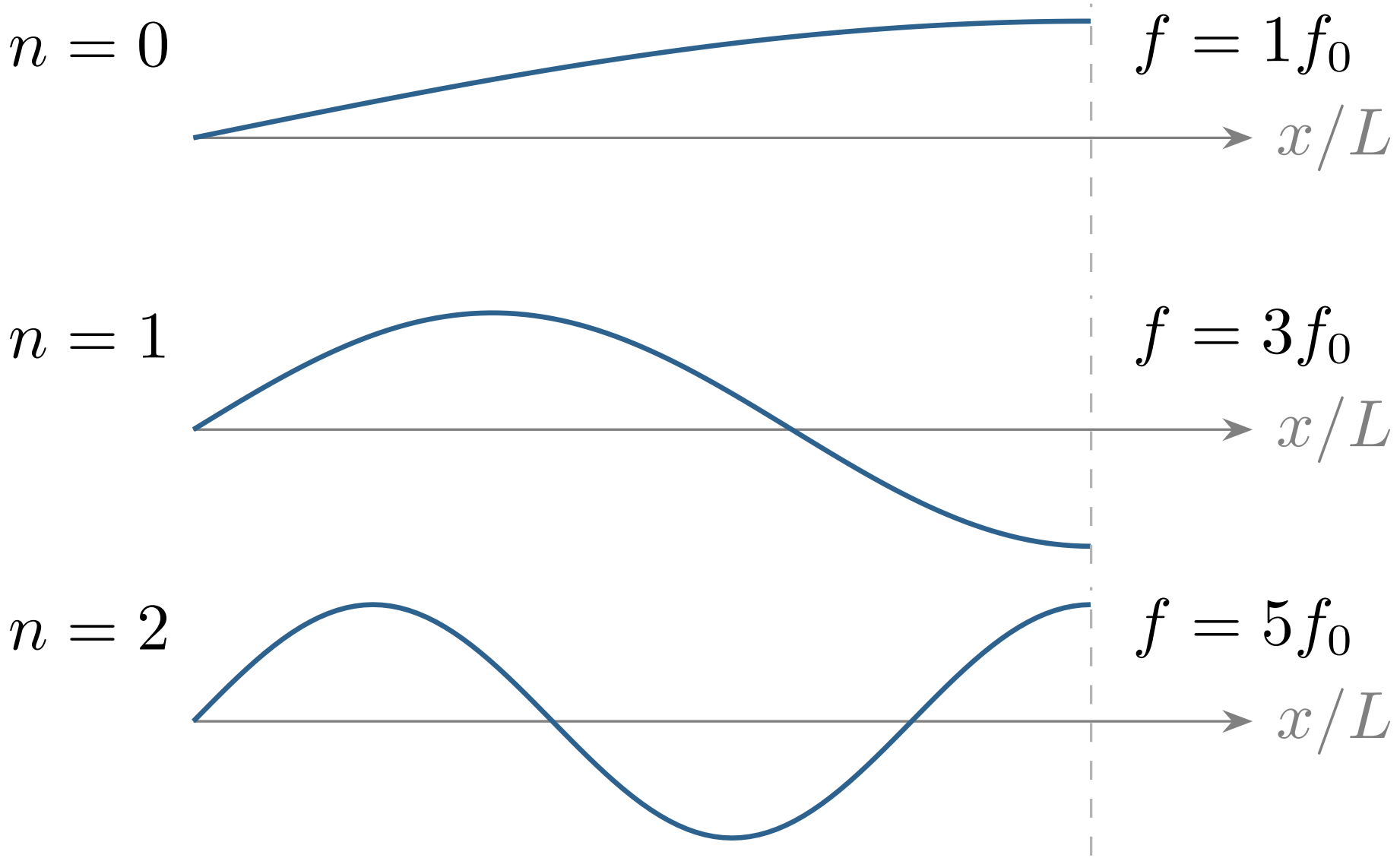
透明侧壁示意；声波沿管轴传播

封闭端　　开口端





表示空间振型；实际两物理量须带各自时间相位



闭端均为节点，开端均满足 $\partial_x \xi = 0$